Float Mounted Depth Sounder (S.O.O. 6295) – General data

1. General

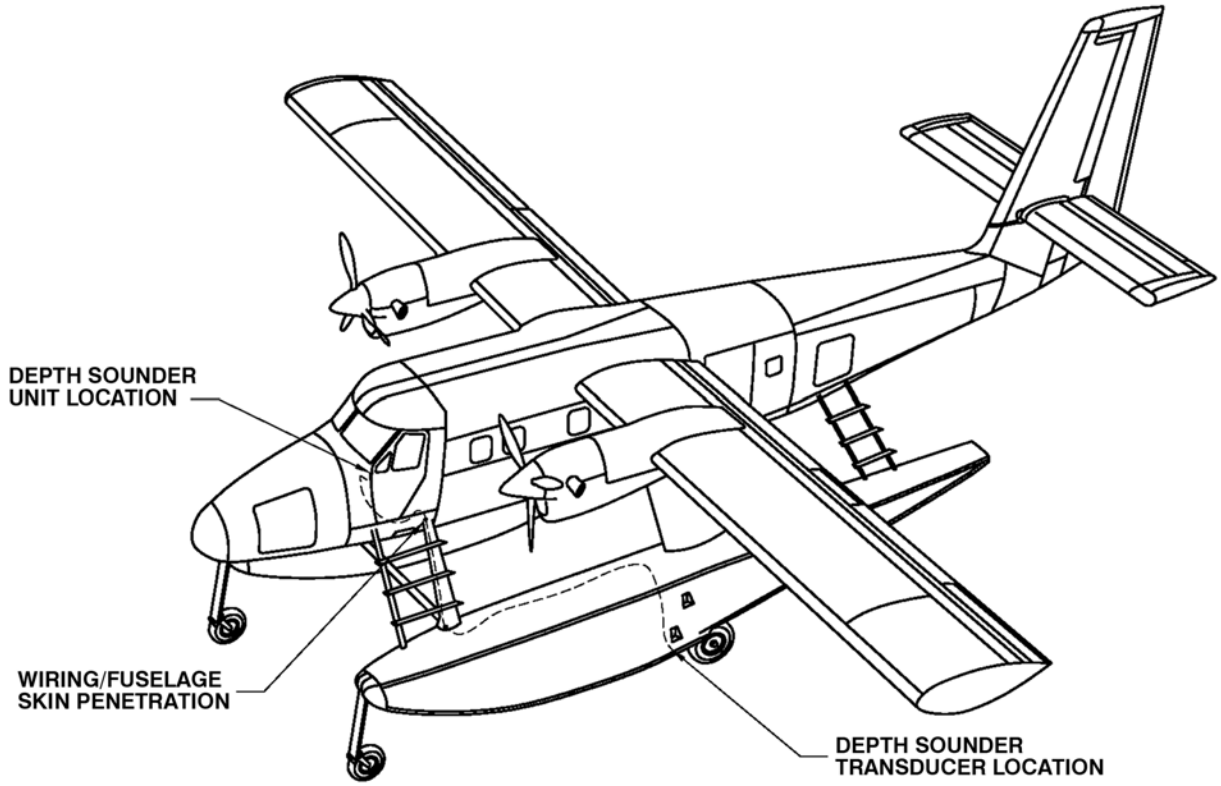
Provisions are made for S.O.O. 6295 Float Mounted Depth Sounder. The depth sounder provides a water depth measurement when the aircraft is operating on Wipaire 13000 amphibious floats or Wipaire 13000 straight floats. The depth sounder is not to be used during flight.

Refer to manufacturer's information regarding maintenance, servicing and repair of Depth Sounder.

2. Description

An illuminated Humminbird depth sounder is located on the pilot's sub panel to the right of the switch panel. The depth sounder provides digital depth reading in feet, meters or fathoms. The depth sounder may also be configured for a shallow water and deep water alarm. Power to the depth sounder is supplied by the 14V power converter through a 1 AMP circuit breaker, connected to the 'line side' of the 14V OUTPUT circuit breaker, located on the avionics circuit breaker panel.

Note Indicated depth measurement depends on bottom hardness, slope and water conditions. Typically, the depth sounder will indicate deeper depths in fresh water than salt water.



Depth Sounder Locations
Figure 1

3. Adjustment/Test

A. Set Measurement Units

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Select DEPTH SOUNDER switch ON.

3) Press the MENU key until the units icon is displayed and flashing.

4) Press the UP or DOWN key to scroll through available units:

- Ft: Feet
- M: Meters
- FA: Fathoms

5) After selection, wait 5 seconds for the unit to return to normal operation. Settings are saved even after depth sounder is powered off.

6) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. Set Shallow Alarm

The Shallow Alarm will sound when the measured water depth is less than the saved setting. If the depth measurement of the water is less than the saved setting, the aural alarm will sound and the alarm icon will flash on the screen. Press any key on the depth sounder unit to mute the aural alarm.

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Select DEPTH SOUNDER switch ON.

3) Press MENU key until the SHALLOW alarm icon is displayed and flashing.

4) Press the UP and DOWN arrow key to change to the desired alarm setting. Default setting is OFF.

5) After selection, wait 5 seconds for the unit to return to normal operation. SHALLOW will be displayed. Settings are saved even after depth sounder is powered off.

6) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

C. Set Deep Alarm

The Deep Alarm will sound when the measured water depth is greater than the saved setting. If the depth measurement of the water is less than the saved setting, the aural alarm will sound and the alarm icon will flash on the screen. Press any key on the depth sounder unit to mute the aural alarm.

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Select DEPTH SOUNDER switch ON.

3) Press MENU key until the DEEP alarm icon is displayed and flashing.

4) Press the UP or DOWN arrow key to change to the desired alarm setting. Default setting is OFF.

Note The minimum Deep Alarm setting cannot equal or drop below the saved Shallow Alarm setting.

5) After selection, wait 5 seconds for the unit to return to normal operation. The DEEP will be displayed. Settings are saved even after depth sounder is powered off.

6) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

D. Set Offset

1) Keel

The Offset function adjusts the digital readout to display reading from the keel (in lieu of the transducer location). Measure the vertical distance between the face of the transducer and keel of the float.

a) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

b) Select DEPTH SOUNDER switch ON.

c) Press the MENU key until SET OFF SET is displayed and flashing on the screen.

d) Press the DOWN arrow key to enter the measurement as a negative (–) number.

e) After the offset is entered, wait 5 seconds for the unit to return to normal operation. An offset icon will be visible on the display. Settings are saved even after the depth sounder is powered off.

f) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Waterline

The Offset function adjusts the digital readout to display reading from the waterline (in lieu of the transducer location). Measure the vertical distance between the face of the transducer and waterline.

a) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

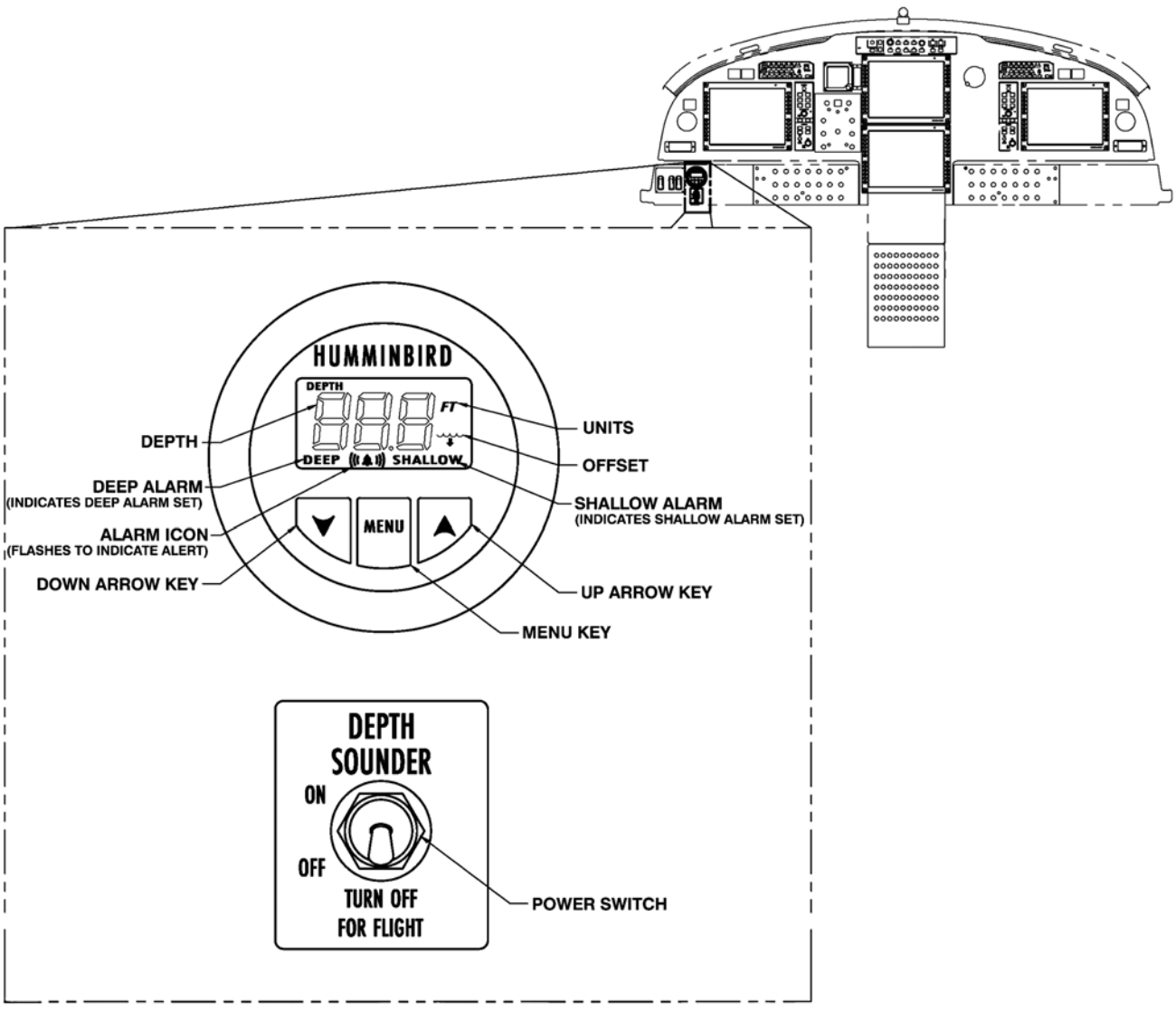
b) Select DEPTH SOUNDER switch ON.

c) Press the MENU key until SET OFF SET is displayed and flashing on the screen.

d) Press the UP arrow key to enter the measurement as a positive (+) number.

e) After the offset is entered, wait 5 seconds for the unit to return to normal operation. An offset icon will be visible on the display. Settings are saved even after the depth sounder is powered off.

f) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).



Depth Sounder Instrument Panel Location
Figure 2

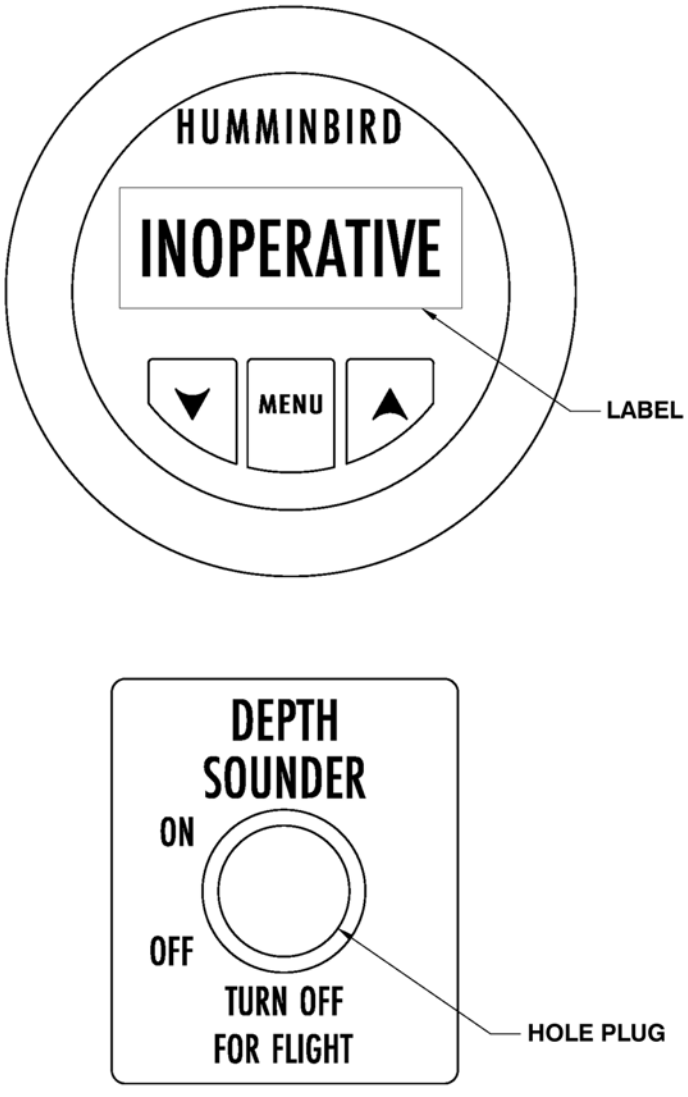
4. Removal

   A. Remove the DEPTH SOUNDER switch and plug hole with .50 inch diameter hole plug. Cap and stow wires.

   B. Remove DEPTH SOUNDER circuit breaker (located on avionics breaker panel) and plug hole with .50 inch diameter hole plug. Cap and stow wires.

   C. Disconnect depth sounder transducer wiring at inline splice located at transducer wire/fuselage penetration. Cap and stow wires.

   D. Fabricate 'INOPERATIVE' label with .25 inch lettering and apply to face of depth sounder display.



Depth Sounder Inoperative Configuration
Figure 3

5. Troubleshooting

Table 1 Troubleshooting

	SYMPTOM	POSSIBLE CAUSE	CHECK THAT	REMEDIAL ACTION
1. (a)	Blank HDR 650 display	Poor connection to power cable.	Condition of depth sounder wiring, splices and connections.	Replace damaged wires Replace damaged splices/connections.
(b)		Incorrectly wired power cable.	'Red' wire is connected to depth sounder switch. 'Black' wire connected to ground.	Correct wiring.
(c)		DEPTH SOUNDER RELAY is open.		Close depth sounder relay.
(d)		Supplied voltage below 10V.	Check output of 14V power converter Voltage should be 14V. Check condition or wiring and connectors.	Replace any damaged wiring. Replace damaged connectors. Replace damaged 14V power converter.
(e)		Damaged depth sounder unit.		Replace HDR 650 with new unit.
2. (a)	No bottom reading on display	Damaged transducer.	Check for visual damage to transducer.	Replace damaged transducer.
(b)		Damaged transducer wiring.	Check condition of transducer wiring in float. Check condition of transducer wiring in aircraft. Check condition of inline terminal module.	Replace damaged transducer wiring. Replace damaged transducer wiring. Replace damaged terminal module.
3.	No continuous depth display in very shallow water	Normal for water depths below one foot.		
4.	Screen fades, Images not sharp	Low voltage supplied to depth sounder.	Refer 1.(a), (b) and (c)	Refer 1.(a), (b) and (c)